# toys10_09-marble-cricket

# Marble Cricket: manufacturing manual

> **Status: design only; nothing has been cut or assembled.**

## 1. What is made

1. Each team gets 2 overs (6 marbles each); drop a marble at the top and read the pocket: runs 0–6 or W (wicket).
2. Two wickets end an innings early. Highest total wins; a tie goes to a one-ball super-over.

## 2. Files and materials

- `toys10/out/marble_cricket_parts.svg` and `.dxf`: **provisional first-pass drawing** (back panel 300 x 400 mm, Galton peg layout, 8 dividers, front panel) generated by `toys10/sims/gen_cut_files.py`; never cut.
- Pocket values left to right chosen by search: see `toys10/docs/results_cricket.md` (wicket chance 14.5 % per ball, mean 10.0 runs per over).
- Material: **6 mm MDF back, 3 mm acrylic front, 6 mm dowel pegs, spacer strips 15 mm (bought-in)**; layout/sheet: 940 x 450 mm layout.

Cost per set (about 2,000 units, estimate):

| Item | Rs |
|---|---|
| Laser-cut back + front + pockets | 120 |
| Pegs ×55 | 20 |
| Marbles ×10 | 15 |
| Levelling screw + feet | 12 |
| Score pad + box | 55 |
| Assembly | 40 |

## 3. Making it

Follow `manuals/STANDARD_PROCEDURES.md` section H (laser cutting) for kerf tests and finishing, then:

1. Cut the back panel; drill/verify 55 peg holes (6.2 mm) and the 16 mm feed hole; glue 6 mm dowel pegs.
2. Glue the 8 pocket dividers at the 9 pocket boundaries and print/paint the pocket values from the results file.
3. Fit 15 mm spacer strips round the edge and the front acrylic panel (marble 12 mm must roll freely, not jam between back and front).
4. Add the levelling screw, feet and score pad; run a **100-marble calibration drop** and compare pocket counts to the binomial expectation: if biased (likely), shift the feed hole or adjust the peg rows.
5. Print the score sheet and rules.

## 4. Inspection (sample per batch)

- 100-marble drop: pocket distribution within the expected binomial band; no marble jams.
- All pockets readable; no sharp edges.
- Wicket rate about 14.5 % per ball in 200 drops (tolerance +-4 %).

Record on `manuals/templates/eol_test_sheet.md`.

## 5. Packaging and compliance

Toys QCO / IS 9873 applicability and testing; age grade and small-parts warnings; Legal Metrology labelling; paint/ink heavy-metal report; no licensed names or logos.

Add the age grade, small-parts warning, Legal Metrology marks and a printed instruction leaflet.

## 6. Risks and limits

- Peg-board bias (calibration test is mandatory).
- Do not use team/league names (IP).
- Marbles: small-parts warning.

Peg-board bias; cricket naming/IP (do not use team or league names).

`docs/results_cricket.md`: wicket chance per ball 14.5 %, mean 10.0 runs per over (10th–90th percentile 4–16), 2-over match tie rate 4.1 %. Pocket values found by search under rules (centre pockets cheap, 6 only on side pockets). Not verified: real marble bounce (likely biased; a 100-marble calibration test is part of the prototype).